



Government of Pakistan
National Vocational & Technical Training Commission (NAVTTTC)

Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



Roll no: 1153541

Name: Muhammad Abubakar Fakhar

CNIC: 3110165887061

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

Test Center Name: DevCastle Training institute |
Software house Bahawalnagar Campus

Examiner Name: Sohail Murtaza /

Signatures

Candidate Signature

Examiner Signature & Stamp

Principal Signature

Instructions:

Please read instructions carefully before attempting paper:

- 1) All questions are compulsory.
- 2) The question paper consists of 21 questions divided into two sections A & B. Section A comprise of 20 MCQs of 1 marks each. Section B comprises of 1 practical task having 80 marks.
- 3) Mobile Phone not allowed.
- 4) Use separate paper for rough work.

Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- When using a cloud service, what is the URL or network address that serves as the entry point for accessing the service's features and functionalities?

- (a) API Key
- (b) Default Endpoint
- (c) Service Manifest
- (d) Authentication Token

2- Which Numpy function is used to perform element-wise addition of two arrays?

- (a) numpy.add()
- (b) numpy.append()
- (c) numpy.sum()
- (d) numpy.plus()

3- In supervised machine learning, what is the primary task of the model?

- (a) To generate new data samples
- (b) To preprocess the data
- (c) To cluster similar data points
- (d) To learn from labeled training data to make predictions on new, unseen data

4- If you're developing an AI solution that involves extracting meaningful insights from unstructured text data, identifying sentiment, entities, and key phrases, which cloud service offers a comprehensive Text Analytics API?

- (a) Azure Cognitive Services
- (b) IBM Watson
- (c) Google Cloud AI
- (d) Amazon SageMaker

5- Which deep learning framework is developed by Google and provides a comprehensive suite of tools and libraries for building and training neural networks?

- (a) Keras
- (b) PyTorch
- (c) TensorFlow
- (d) Theano

6- When developing an application that involves extracting insights, sentiment, and key phrases from text data, which Azure service provides text analytics capabilities for language detection, sentiment analysis, and named entity recognition?

- (a) Azure Content Moderator
- (b) Azure Speech Service
- (c) Azure Text Analytics
- (d) Azure Translator

7- Which keyword is used in Python to introduce an alternative condition if the initial condition is false?

- (a) Elself
- (b) Otherwise
- (c) Else
- (d) Unless

8- Which data structure is commonly used in Pandas to store tabular data?

- (a) Arrays
- (b) Lists
- (c) DataFrames
- (d) Sets

9- In Pandas, what operation allows you to group a DataFrame based on a column or columns and then perform aggregate functions on each group?

- (a) GroupBy
- (b) Aggregate
- (c) Group
- (d) Combine

10- In machine learning, what is the process of selecting a subset of relevant features from the original set of features to improve model performance and reduce complexity?

- (a) Feature extraction
- (b) Feature selection
- (c) NER tagging
- (d) Tokenization

11- Which type of words are commonly removed from text during pre-processing due to their high frequency and lack of significant meaning in most contexts?

- (a) Verbs
- (b) Stopwords
- (c) Adjectives
- (d) Punctuation

12- In neural networks, what are the two main steps involved in the training process, where the first step involves computing the output of the network given the input data, and the second step involves updating the network's weights based on the error?

- (a) Data Preprocessing and Feature Selection
- (b) Forward and Backward Passes
- (c) Hyperparameter Tuning and Cross-Validation
- (d) Model Compilation and Evaluation

13- What is an object in object-oriented programming?

- (a) A block of code that is executed once
- (b) A function that defines a class
- (c) An instance of a class with its own properties and methods
- (d) A value that can be changed during runtime

14- Which Pandas plotting method is used to create a horizontal bar plot?

- (a) `df.plot.hist()`
- (b) `df.plot.barh()`
- (c) `df.plot.bar()`
- (d) `df.plot.line()`

15- Which Numpy array attribute returns the number of dimensions (axes) in the array?

- (a) `shape`
- (b) `size`
- (c) `dtype`
- (d) `ndim`

16- What is the advantage of using functions in programming?

- (a) They allow you to define new data types
- (b) They help you optimize memory usage
- (c) They allow you to create GUI interfaces
- (d) They allow you to reuse code and improve code organization

17- In Pandas, which operation is used to perform element-wise addition of two DataFrames with matching indexes and columns?

- (a) df.concat()
- (b) df.add()
- (c) df.combine()
- (d) df.merge()

18- In natural language processing, what is the process of identifying and classifying named entities in text, such as names of people, locations, organizations, and dates?

- (a) NER tagging
- (b) POS tagging
- (c) Stemming
- (d) Tokenization

19- In cloud services, what are used as credentials to securely authenticate and access resources, and are often required to be protected and rotated regularly to enhance security?

- (a) API keys
- (b) Account keys
- (c) Access tokens
- (d) Encryption keys

20- In Python, what is the purpose of file handling?

- (a) To delete unnecessary files
- (b) To control the display of files in the terminal
- (c) To create backups of important files
- (d) To read from and write to files on disk

__The End__



Government of Pakistan
National Vocational & Technical Training Commission (NAVTTTC)

Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



Roll no: 1153549

Name: Afeera Sharif

CNIC: 3110157368492

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

Test Center Name: DevCastle Training institute |
Software house Bahawalnagar Campus

Examiner Name: Sohail Murtaza /

Signatures

Candidate Signature

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- 4) Use separate paper for rough work.

Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- In the classical machine learning pipeline, what step involves transforming raw data into a suitable format for analysis?

- (a) Feature crafting
- (b) Data preprocessing
- (c) Modeling
- (d) Testing and validation

2- What is the primary application of natural language processing (NLP) solutions?

- (a) Image analysis
- (b) Video compression
- (c) Text analysis and understanding
- (d) Audio processing

3- In Pandas, which data structure is used to store one-dimensional labeled data, similar to a column in a spreadsheet?

- (a) List
- (b) DataFrame
- (c) Array
- (d) Series

4- When planning for an AI solution that aligns with Responsible AI principles, which practice involves ensuring transparency and explainability of AI models and decisions, enabling users to understand the reasoning behind predictions?

- (a) Model Accuracy
- (b) Explainability and Transparency
- (c) Performance Metrics
- (d) Data Collection

5- Which measure provides an indication of how spread out the values in a dataset are?

- (a) Z-Score
- (b) Coefficient of Variation
- (c) Variance
- (d) Standard Deviation

6- Seaborn is a Python data visualization library based on which other library?

- (a) Scikit-learn
- (b) Matplotlib
- (c) Pandas
- (d) Numpy

7- What is dynamic polymorphism in object-oriented programming?

- (a) A technique to combine multiple classes into one
- (b) The ability of a subclass to override methods of its base class
- (c) A way to prevent inheritance between classes
- (d) The process of defining new data types

8- In supervised machine learning, which component represents the function that the model is trying to learn and approximate?

- (a) Optimizer
- (b) Hypothesis
- (c) Cost function
- (d) Labeled data

9- What type of data consists of numerical values that represent quantities?

- (a) Binary data
- (b) Categorical data
- (c) Qualitative data
- (d) Quantitative data

10- In supervised machine learning, what is the primary task of the model?

- (a) To generate new data samples
- (b) To preprocess the data
- (c) To cluster similar data points
- (d) To learn from labeled training data to make predictions on new, unseen data

11- What does broadcasting in Numpy allow you to do?

- (a) Convert arrays to different data types
- (b) Perform element-wise operations on arrays with different shapes
- (c) Combine arrays into a single array
- (d) Resize arrays to match the largest array

12- When developing a solution that automatically detects and filters inappropriate or offensive content from text, images, and videos, which Azure service provides the capability to moderate and classify content to ensure it meets appropriate standards?

- (a) Azure Cognitive Services
- (b) Azure Content Moderator
- (c) Azure Machine Learning
- (d) Azure Computer Vision

13- What is the purpose of name mangling in Python classes?

- (a) Obfuscating code
- (b) Creating unique variable names
- (c) Improving code performance
- (d) Preventing access to class attributes

14- Which command is used to change the permissions of a file in Unix-like systems?

- (a) touch
- (b) kill
- (c) chmod
- (d) ps

15- Which Seaborn function is used to create a histogram with a kernel density estimate (KDE) plot?

- (a) sns.distplot()
- (b) sns.histplot()
- (c) sns.kdeplot()
- (d) sns.plot.hist()

16- In machine learning, what term refers to a situation where one class has significantly more instances than other classes in a dataset?

- (a) Overfitting
- (b) Dataset augmentation
- (c) Underfitting
- (d) Dataset imbalance

17- Which Numpy function is used to create a copy of an array?

- (a) numpy.clone()
- (b) numpy.copy()
- (c) numpy.duplicate()
- (d) numpy.replicate()

18- In the Bag-of-Words (BoW) model, how is a text represented as a feature vector for NLP tasks?

- (a) By using lemmatization
- (b) By counting the frequency of words in the text
- (c) By removing stopwords
- (d) By converting words to lowercase

19- Which keyword is used in Python to introduce an alternative condition if the initial condition is false?

- (a) Elself
- (b) Otherwise
- (c) Else
- (d) Unless

20- What method is used to select elements from a Numpy array based on a condition?

- (a) conditional_selection()
- (b) filter()
- (c) logical_select()
- (d) where()

__The End__



Roll no: 1153547

Name: ASADULLAH

CNIC: 3660114393477

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

Test Center Name: DevCastle Training institute |
Software house Bahawalnagar Campus

Examiner Name: Sohail Murtaza /

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Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- In languages without explicit spaces between words, what is the process of dividing a sequence of characters into individual words called?

- (a) Sentence tokenization
- (b) Part-of-speech tagging
- (c) Text normalization
- (d) Word segmentation

2- Which Seaborn function is used to create a bar plot of categorical data?

- (a) sns.catplot()
- (b) sns.barplot()
- (c) sns.scatterplot()
- (d) sns.lineplot()

3- Which command is used to display the contents of a file in a Unix-like operating system?

- (a) read
- (b) view
- (c) cat
- (d) open

4- When developing a solution that automatically detects and filters inappropriate or offensive content from text, images, and videos, which Azure service provides the capability to moderate and classify content to ensure it meets appropriate standards?

- (a) Azure Cognitive Services
- (b) Azure Content Moderator
- (c) Azure Machine Learning
- (d) Azure Computer Vision

5- When creating an inclusive and accessible application that helps users read content more easily, which Azure service provides features like text-to-speech, translation, and focus mode, enhancing readability for users with varying abilities?

- (a) Azure Personalizer
- (b) Azure Content Moderator
- (c) Azure Text Analytics
- (d) Azure Immersive Reader

6- Which keyword is used in Python to introduce an alternative condition if the initial condition is false?

- (a) Elself
- (b) Otherwise
- (c) Else
- (d) Unless

7- In the classical machine learning pipeline, what step involves transforming raw data into a suitable format for analysis?

- (a) Feature crafting
- (b) Data preprocessing
- (c) Modeling
- (d) Testing and validation

8- Which library is commonly used for natural language processing tasks such as tokenization, stemming, and part-of-speech tagging?

- (a) Matplotlib
- (b) SpaCy
- (c) Scikit-learn
- (d) NLTK

9- In logistic regression, what type of output does the model produce?

- (a) Probability scores
- (b) Continuous numerical values
- (c) Categorical labels
- (d) Clusters of data points

10- What field of study focuses on enabling computers to understand, interpret, and generate human language?

- (a) Computer Vision
- (b) Data Science
- (c) Natural Language Processing
- (d) Machine Learning

11- When creating a new resource in Microsoft Azure, which web-based interface provides a graphical user interface (GUI) that allows you to select and configure various Azure services and resources?

- (a) Azure PowerShell
- (b) Azure Portal
- (c) Azure CLI
- (d) Azure SDK

12- In statistics, which term describes a variable that can take on distinct, separate values?

- (a) Quantitative
- (b) Qualitative
- (c) Discrete
- (d) Continuous

13- In object-oriented programming, what is an abstract method?

- (a) A method that can't be called
- (b) A method that is automatically generated
- (c) A method that has no implementation
- (d) A method that can't be overridden

14- In machine learning, what term refers to a situation where one class has significantly more instances than other classes in a dataset?

- (a) Overfitting
- (b) Dataset augmentation
- (c) Underfitting
- (d) Dataset imbalance

15- In Pandas, which method is used to save a DataFrame to a CSV file?

- (a) df.write_csv()
- (b) df.export_csv()
- (c) df.to_csv()
- (d) df.save_csv()

16- What is hierarchical inheritance in OOP?

- (a) Inheriting from a single parent class
- (b) Inheriting from a grandparent class
- (c) Inheriting from multiple child classes
- (d) Inheriting from multiple parent classes

17- If you're building a speech solution that involves converting spoken language into text, performing text-to-speech synthesis, and enabling speaker recognition, which Azure Cognitive Services offering is best suited for these tasks?

- (a) Azure Text Analytics
- (b) Azure Language Understanding
- (c) Azure Speech Service
- (d) Azure Translator

18- What is the purpose of selecting the appropriate Microsoft Azure AI service for a solution?

- (a) To simplify the code development process
- (b) To ensure the responsible use of AI technologies
- (c) To maximize profits from AI solutions
- (d) To showcase various AI services to clients

19- When developing an application that involves extracting insights, sentiment, and key phrases from text data, which Azure service provides text analytics capabilities for language detection, sentiment analysis, and named entity recognition?

- (a) Azure Content Moderator
- (b) Azure Speech Service
- (c) Azure Text Analytics
- (d) Azure Translator

20- Which deep learning framework is developed by Google and provides a comprehensive suite of tools and libraries for building and training neural networks?

- (a) Keras
- (b) PyTorch
- (c) TensorFlow
- (d) Theano

__The End__



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National Vocational & Technical Training Commission (NAVTTTC)

Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



Roll no: 1153545

Name: Saba Zafar

CNIC: 3220289618114

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

Test Center Name: DevCastle Training institute |
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Examiner Name: Sohail Murtaza /

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Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- How can you convert a user's input into a numerical data type in programming?

- (a) By performing a bitwise operation
- (b) Through the input() function and type casting
- (c) Using the stringify() function
- (d) By using the floatify() method

2- In object-oriented programming, what are instance variables?

- (a) Variables that are shared among all instances of a class
- (b) Variables that store data unique to each instance of a class
- (c) Variables that belong to the class itself
- (d) Variables used for method definition

3- Which Numpy function is used to create a copy of an array?

- (a) numpy.clone()
- (b) numpy.copy()
- (c) numpy.duplicate()
- (d) numpy.replicate()

4- Which term in NLP refers to a curve that plots the performance of a model as a function of a certain parameter?

- (a) Discourse curve
- (b) Pragmatics curve
- (c) Learning curve
- (d) Syntax curve

5- What does the term "probability" refer to in statistics?

- (a) The ratio of successful outcomes to the total possible outcomes
- (b) The number of trials in an experiment
- (c) The certainty of an event occurring
- (d) The number of times an event occurs

6- When planning for an AI solution that aligns with Responsible AI principles, which practice involves ensuring transparency and explainability of AI models and decisions, enabling users to understand the reasoning behind predictions?

- (a) Model Accuracy
- (b) Explainability and Transparency
- (c) Performance Metrics
- (d) Data Collection

7- In logistic regression, what type of output does the model produce?

- (a) Probability scores
- (b) Continuous numerical values
- (c) Categorical labels
- (d) Clusters of data points

8- When building a solution that involves converting spoken language into written text and performing speech-to-text transcription, which Azure service offers automatic speech recognition capabilities for converting spoken language into text data?

- (a) Azure Translator
- (b) Azure Text Analytics
- (c) Azure Speech Service
- (d) Azure Content Moderator

9- In machine learning, which algorithm is used for both classification and regression tasks and aims to find the hyperplane that best separates data points of different classes while maximizing the margin?

- (a) Linear Regression
- (b) Support Vector Machine (SVM)
- (c) Decision Tree
- (d) K-Means Clustering

10- In supervised machine learning, what is the primary task of the model?

- (a) To generate new data samples
- (b) To preprocess the data
- (c) To cluster similar data points
- (d) To learn from labeled training data to make predictions on new, unseen data

11- In Linux, which command is used to update the package list?

- (a) apt-get
- (b) apt-upgrade
- (c) apt-list
- (d) apt-update

12- In machine learning, which task involves predicting discrete class labels as output?

- (a) Dimensionality reduction
- (b) Classification
- (c) Regression
- (d) Clustering

13- Which type of loop is suitable when the number of iterations is known beforehand?

- (a) While loop
- (b) Do-while loop
- (c) For loop
- (d) Infinite loop

14- What type of data consists of numerical values that represent quantities?

- (a) Binary data
- (b) Categorical data
- (c) Qualitative data
- (d) Quantitative data

15- In NLP, what is the process of converting text to a consistent format, such as converting all characters to lowercase and removing special characters?

- (a) Stemming
- (b) Part-of-speech tagging
- (c) Tokenization
- (d) Text normalization

16- When integrating Azure AI services into a continuous integration/continuous deployment (CI/CD) pipeline, which practice involves automating the deployment of AI models and services to production environments, ensuring consistent and efficient deployment p

- (a) CI/CD Integration
- (b) Data Collection
- (c) Model Training
- (d) Feature Engineering

17- In natural language processing, what is the process of identifying and classifying named entities in text, such as names of people, locations, organizations, and dates?

- (a) NER tagging
- (b) POS tagging
- (c) Stemming
- (d) Tokenization

18- In Pandas, which type of join combines rows from two DataFrames based on common columns, keeping only the matching rows?

- (a) Left join
- (b) Right join
- (c) Inner join
- (d) Outer join

19- In machine learning, what term refers to a situation where one class has significantly more instances than other classes in a dataset?

- (a) Overfitting
- (b) Dataset augmentation
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20- Which Numpy function is used to perform element-wise addition of two arrays?

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- (d) numpy.plus()

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Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



Roll no: 1153546

Name: Tuba Shakeel

CNIC: 3110164503982

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

Test Center Name: DevCastle Training institute |
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Examiner Name: Sohail Murtaza /

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- (a) API Key
- (b) Default Endpoint
- (c) Service Manifest
- (d) Authentication Token

2- What does polymorphism in OOP refer to?

- (a) Overriding parent class methods
- (b) Allowing operators to work with different data types objects as instances of a common superclass
- (c) Creating complex class hierarchies
- (d) Treating different class objects as instances of a common superclass

3- What is the purpose of software installation tools like Anaconda and VSCode in AI development?

- (a) To demonstrate origami techniques
- (b) To install and manage necessary programming environments
- (c) To showcase gardening skills
- (d) To create virtual reality experiences

4- What is an inner/nested class in OOP?

- (a) A class defined within another class
- (b) A class that can only be instantiated once
- (c) A class that inherits from another class
- (d) A class with no methods or attributes

5- In object-oriented programming, what are instance variables?

- (a) Variables that are shared among all instances of a class
- (b) Variables that store data unique to each instance of a class
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- (a) A block of code that is executed once
- (b) A function that defines a class
- (c) An instance of a class with its own properties and methods
- (d) A value that can be changed during runtime

9- What is a lexical database in NLP that provides a semantic network of words and their relationships, including synonyms and hypernyms?

- (a) NER tagging
- (b) Stemming
- (c) POS tagging
- (d) WordNet

10- How can you convert a user's input into a numerical data type in programming?

- (a) By performing a bitwise operation
- (b) Through the input() function and type casting
- (c) Using the stringify() function
- (d) By using the floatify() method

11- In machine learning, what type of problem involves predicting a continuous numerical output?

- (a) Regression
- (b) Classification
- (c) Reinforcement learning
- (d) Clustering

12- When developing an application that requires translating text from one language to another, which Azure service provides language translation capabilities and supports a wide range of languages for translation?

- (a) Azure Text Analytics
- (b) Azure Content Moderator
- (c) Azure Speech Service
- (d) Azure Translator

13- In Pandas, which data structure is used to store one-dimensional labeled data, similar to a column in a spreadsheet?

- (a) List
- (b) DataFrame
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- (d) Series

14- In cloud services, what are used as credentials to securely authenticate and access resources, and are often required to be protected and rotated regularly to enhance security?

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- (a) Azure Personalizer
- (b) Azure Content Moderator
- (c) Azure Text Analytics
- (d) Azure Immersive Reader

16- Which measure provides an indication of how spread out the values in a dataset are?

- (a) Z-Score
- (b) Coefficient of Variation
- (c) Variance
- (d) Standard Deviation

17- Which evaluation metric is commonly used for binary classification problems and calculates the ratio of correctly predicted positive observations to the total predicted positives?

- (a) Accuracy
- (b) Precision
- (c) F1-score
- (d) Recall

18- In Microsoft Azure, which service provides a platform for building, training, and deploying machine learning models, enabling you to create AI-powered solutions using a variety of tools and frameworks?

- (a) Azure Machine Learning
- (b) Azure AI Builder
- (c) Azure Data Factory
- (d) Azure Cognitive Services

19- In sequence models, which type of recurrent neural network (RNN) architecture processes input sequences both in forward and backward directions, capturing context from both past and future elements?

- (a) Bi-Directional LSTM/RNN
- (b) CNN
- (c) Uni-Directional LSTM/RNN
- (d) Transformer

20- In natural language processing, which model aims to predict a target word based on its surrounding context words, and is often used for learning dense word representations?

- (a) Continuous Bag-of-Words (CBOW)
- (b) Named Entity Recognition
- (c) Part-of-Speech Tagging
- (d) Skip-gram

__The End__



Government of Pakistan
National Vocational & Technical Training Commission (NAVTTTC)

Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



Roll no: 1153531

Name: husnain bashir

CNIC: 3110134430337

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

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- 3) Mobile Phone not allowed.
- 4) Use separate paper for rough work.

Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- How can you create a Numpy array from a Python list?

- (a) numpy.create()
- (b) numpy.array()
- (c) numpy.from_list()
- (d) numpy.list()

2- In sequence models, which type of recurrent neural network (RNN) architecture processes input sequences both in forward and backward directions, capturing context from both past and future elements?

- (a) Bi-Directional LSTM/RNN
- (b) CNN
- (c) Uni-Directional LSTM/RNN
- (d) Transformer

3- When planning for an AI solution that aligns with Responsible AI principles, which practice involves ensuring transparency and explainability of AI models and decisions, enabling users to understand the reasoning behind predictions?

- (a) Model Accuracy
- (b) Explainability and Transparency
- (c) Performance Metrics
- (d) Data Collection

4- Which Seaborn function is used to create a bar plot of categorical data?

- (a) sns.catplot()
- (b) sns.barplot()
- (c) sns.scatterplot()
- (d) sns.lineplot()

5- In the Bag-of-Words (BoW) model, how is a text represented as a feature vector for NLP tasks?

- (a) By using lemmatization
- (b) By counting the frequency of words in the text
- (c) By removing stopwords
- (d) By converting words to lowercase

6- What is a common step in data pre-processing for NLP that involves converting all characters to lowercase?

- (a) Lemmatization
- (b) Stopword removal
- (c) Tokenization
- (d) Lowercasing

7- Which command is used to display the contents of a directory in the terminal?

- (a) ls
- (b) display
- (c) dir
- (d) show

8- What type of data consists of numerical values that represent quantities?

- (a) Binary data
- (b) Categorical data
- (c) Qualitative data
- (d) Quantitative data

9- In object-oriented programming, what is multilevel inheritance?

- (a) Inheriting from multiple parent classes
- (b) Inheriting from a child class
- (c) Inheriting from a grandparent class
- (d) Inheriting from a single parent class

10- In programming, what is the term for a function defined inside another function?

- (a) Nested function
- (b) Child function
- (c) Inner function
- (d) Subfunction

11- Which Seaborn function is used to create a histogram with a kernel density estimate (KDE) plot?

- (a) `sns.distplot()`
- (b) `sns.histplot()`
- (c) `sns.kdeplot()`
- (d) `sns.plot.hist()`

12- Which symbol is used in a regular expression to match any single character?

- (a) `^`
- (b) `*`
- (c) `.`
- (d) `?`

13- Which type of words are commonly removed from text during pre-processing due to their high frequency and lack of significant meaning in most contexts?

- (a) Verbs
- (b) Stopwords
- (c) Adjectives
- (d) Punctuation

14- When developing an application that requires custom image classification and object detection, which Azure Cognitive Service provides a user-friendly platform to train and deploy custom models for various visual recognition tasks?

- (a) Azure Computer Vision
- (b) Custom Vision
- (c) Azure Content Moderator
- (d) Azure Video Indexer

15- How can you convert a user's input into a numerical data type in programming?

- (a) By performing a bitwise operation
- (b) Through the `input()` function and type casting
- (c) Using the `stringify()` function
- (d) By using the `floatify()` method

16- What is the purpose of list comprehension in Python?

- (a) To create complex nested lists
- (b) To manipulate and filter lists using a concise syntax
- (c) To write long and verbose loops
- (d) To generate random numbers

17- Which ensemble method combines weak learners sequentially, where each subsequent learner focuses more on misclassified data from the previous ones, aiming to create a strong learner?

- (a) K-Means Clustering
- (b) Random Forest
- (c) Bagging
- (d) Boosting

18- What is hierarchical inheritance in OOP?

- (a) Inheriting from a single parent class
- (b) Inheriting from a grandparent class
- (c) Inheriting from multiple child classes
- (d) Inheriting from multiple parent classes

19- In machine learning, what type of problem involves predicting a continuous numerical output?

- (a) Regression
- (b) Classification
- (c) Reinforcement learning
- (d) Clustering

20- Which measure indicates how many standard deviations a data point is away from the mean?

- (a) Quartile
- (b) Z-Score
- (c) Range
- (d) Percentile

__The End__



Government of Pakistan
National Vocational & Technical Training Commission (NAVTTTC)

Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



Roll no: 1153548

Name: Farea Rafiq

CNIC: 3660193403776

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

Test Center Name: DevCastle Training institute |
Software house Bahawalnagar Campus

Examiner Name: Sohail Murtaza /

Signatures

Candidate Signature

Examiner Signature & Stamp

Principal Signature

Instructions:

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- 3) Mobile Phone not allowed.
- 4) Use separate paper for rough work.

Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- What is the primary application of Convolutional Neural Networks (CNNs)?

- (a) Tabular data analysis
- (b) Text classification
- (c) Image and video analysis
- (d) Speech recognition

2- Which data structure is used to store an unordered collection of unique elements in Python?

- (a) Arrays
- (b) Sets
- (c) Lists
- (d) Tuples

3- In deep learning, what type of neural network architecture consists of multiple layers of interconnected neurons, including input, hidden, and output layers, where information flows only in one direction from input to output?

- (a) Recurrent Neural Network (RNN)
- (b) Long Short-Term Memory (LSTM)
- (c) MLP Feed Forward Neural Network
- (d) Convolutional Neural Network (CNN)

4- What is the purpose of "magic functions" or "dunder methods" in Python?

- (a) Enforcing strict naming conventions
- (b) Defining private methods in a class
- (c) Creating unique variable names
- (d) Overloading operators and providing special behavior

5- What is the purpose of software installation tools like Anaconda and VSCode in AI development?

- (a) To demonstrate origami techniques
- (b) To install and manage necessary programming environments
- (c) To showcase gardening skills
- (d) To create virtual reality experiences

6- In machine learning, which algorithm is used for both classification and regression tasks and aims to find the hyperplane that best separates data points of different classes while maximizing the margin?

- (a) Linear Regression
- (b) Support Vector Machine (SVM)
- (c) Decision Tree
- (d) K-Means Clustering

7- In object-oriented programming, what is an abstract method?

- (a) A method that can't be called
- (b) A method that is automatically generated
- (c) A method that has no implementation
- (d) A method that can't be overridden

8- Which technique in natural language processing is used to convert words into dense vector representations, capturing semantic meanings and relationships between words?

- (a) Word2Vec
- (b) Named Entity Recognition
- (c) Bag-of-Words
- (d) POS tagging

9- When managing authentication for a cloud resource, which approach involves configuring and using identity and access management (IAM) policies to control who can access and perform actions on the resource?

- (a) IAM Policies
- (b) OAuth Tokens
- (c) API Keys
- (d) Encryption Keys

10- If you're developing an AI solution that involves extracting meaningful insights from unstructured text data, identifying sentiment, entities, and key phrases, which cloud service offers a comprehensive Text Analytics API?

- (a) Azure Cognitive Services
- (b) IBM Watson
- (c) Google Cloud AI
- (d) Amazon SageMaker

11- Which measure of central tendency is most affected by outliers?

- (a) Standard Deviation
- (b) Mean
- (c) Median
- (d) Mode

12- Which measure indicates how many standard deviations a data point is away from the mean?

- (a) Quartile
- (b) Z-Score
- (c) Range
- (d) Percentile

13- In natural language processing, what is the process of reducing a word to its base or root form called, by removing suffixes or prefixes?

- (a) Segmentation
- (b) Stemming
- (c) Lemmatization
- (d) Tokenization

14- When deploying a machine learning model as a web service in Microsoft Azure, which service allows you to create, deploy, and manage RESTful APIs for your trained models, making them accessible for predictions?

- (a) Azure Cognitive Services
- (b) Azure Functions
- (c) Azure Logic Apps
- (d) Azure Machine Learning Service

15- What are universal array functions in Numpy?

- (a) Functions that perform matrix multiplication
- (b) Functions that operate element-wise on arrays
- (c) Functions that only work on 1D arrays
- (d) Functions that require external libraries

16- What does the term "probability" refer to in statistics?

- (a) The ratio of successful outcomes to the total possible outcomes
- (b) The number of trials in an experiment
- (c) The certainty of an event occurring
- (d) The number of times an event occurs

17- In Python, what is the purpose of file handling?

- (a) To delete unnecessary files
- (b) To control the display of files in the terminal
- (c) To create backups of important files
- (d) To read from and write to files on disk

18- In machine learning, what term refers to a situation where a model learns the training data too well, capturing noise and leading to poor generalization on new data?

- (a) Bias-Variance Tradeoff
- (b) Regularization
- (c) Underfitting
- (d) Overfitting

19- In statistics, which term describes a variable that can take on distinct, separate values?

- (a) Quantitative
- (b) Qualitative
- (c) Discrete
- (d) Continuous

20- Which Seaborn function is used to create a heatmap to visualize the correlation matrix of a DataFrame?

- (a) `sns.corrplot()`
- (b) `sns.matrixplot()`
- (c) `sns.correlationplot()`
- (d) `sns.heatmap()`

__The End__



Government of Pakistan
National Vocational & Technical Training Commission (NAVTTTC)

Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



Roll no: 1153544

Name: Abdul Manan

CNIC: 3110150166857

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

Test Center Name: DevCastle Training institute |
Software house Bahawalnagar Campus

Examiner Name: Sohail Murtaza /

Signatures

Candidate Signature

Examiner Signature & Stamp

Principal Signature

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- 3) Mobile Phone not allowed.
- 4) Use separate paper for rough work.

Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- What does Bayesian probability involve?

- (a) Estimating probabilities using random sampling
- (b) Counting the number of occurrences in a dataset
- (c) Calculating probabilities based on prior knowledge and observations
- (d) Using logical operators to find probabilities

2- In object-oriented programming, what are class variables?

- (a) Variables that are shared among all instances of a class
- (b) Variables that belong to the class itself
- (c) Variables that store data unique to each instance of a class
- (d) Variables used for method definition

3- In supervised machine learning, which component represents the function that the model is trying to learn and approximate?

- (a) Optimizer
- (b) Hypothesis
- (c) Cost function
- (d) Labeled data

4- In deep learning, what type of neural network architecture consists of multiple layers of interconnected neurons, including input, hidden, and output layers, where information flows only in one direction from input to output?

- (a) Recurrent Neural Network (RNN)
- (b) Long Short-Term Memory (LSTM)
- (c) MLP Feed Forward Neural Network
- (d) Convolutional Neural Network (CNN)

5- What is the purpose of selecting the appropriate Microsoft Azure AI service for a solution?

- (a) To simplify the code development process
- (b) To ensure the responsible use of AI technologies
- (c) To maximize profits from AI solutions
- (d) To showcase various AI services to clients

6- When integrating Azure AI services into a continuous integration/continuous deployment (CI/CD) pipeline, which practice involves automating the deployment of AI models and services to production environments, ensuring consistent and efficient deployment p

- (a) CI/CD Integration
- (b) Data Collection
- (c) Model Training
- (d) Feature Engineering

7- Which type of words are commonly removed from text during pre-processing due to their high frequency and lack of significant meaning in most contexts?

- (a) Verbs
- (b) Stopwords
- (c) Adjectives
- (d) Punctuation

8- In Pandas, which type of join combines rows from two DataFrames based on common columns, keeping only the matching rows?

- (a) Left join
- (b) Right join
- (c) Inner join
- (d) Outer join

9- In machine learning, what term refers to a situation where a model learns the training data too well, capturing noise and leading to poor generalization on new data?

- (a) Bias-Variance Tradeoff
- (b) Regularization
- (c) Underfitting
- (d) Overfitting

10- In cloud services, what are used as credentials to securely authenticate and access resources, and are often required to be protected and rotated regularly to enhance security?

- (a) API keys
- (b) Account keys
- (c) Access tokens
- (d) Encryption keys

11- Which command is used to display the contents of a directory in the terminal?

- (a) ls
- (b) display
- (c) dir
- (d) show

12- Which measure quantifies the relative variability of a dataset, considering the standard deviation and mean?

- (a) Coefficient of Variation
- (b) Skewness
- (c) Kurtosis
- (d) Variance

13- Which technique in natural language processing is used to convert words into dense vector representations, capturing semantic meanings and relationships between words?

- (a) Word2Vec
- (b) Named Entity Recognition
- (c) Bag-of-Words
- (d) POS tagging

14- When developing an application that requires custom image classification and object detection, which Azure Cognitive Service provides a user-friendly platform to train and deploy custom models for various visual recognition tasks?

- (a) Azure Computer Vision
- (b) Custom Vision
- (c) Azure Content Moderator
- (d) Azure Video Indexer

15- What is dynamic polymorphism in object-oriented programming?

- (a) A technique to combine multiple classes into one
- (b) The ability of a subclass to override methods of its base class
- (c) A way to prevent inheritance between classes
- (d) The process of defining new data types

16- When using a cloud service, what is the URL or network address that serves as the entry point for accessing the service's features and functionalities?

- (a) API Key
- (b) Default Endpoint
- (c) Service Manifest
- (d) Authentication Token

17- In logistic regression, what type of output does the model produce?

- (a) Probability scores
- (b) Continuous numerical values
- (c) Categorical labels
- (d) Clusters of data points

18- When building a solution that involves converting spoken language into written text and performing speech-to-text transcription, which Azure service offers automatic speech recognition capabilities for converting spoken language into text data?

- (a) Azure Translator
- (b) Azure Text Analytics
- (c) Azure Speech Service
- (d) Azure Content Moderator

19- Which Seaborn function is used to create a histogram with a kernel density estimate (KDE) plot?

- (a) sns.distplot()
- (b) sns.histplot()
- (c) sns.kdeplot()
- (d) sns.plot.hist()

20- Which library is commonly used for natural language processing tasks such as tokenization, stemming, and part-of-speech tagging?

- (a) Matplotlib
- (b) SpaCy
- (c) Scikit-learn
- (d) NLTK

__The End__



Government of Pakistan
National Vocational & Technical Training Commission (NAVTTTC)

Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



Roll no: 1153543

Name: Muhammad Babar Ali

CNIC: 3110152858547

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

Test Center Name: DevCastle Training institute |
Software house Bahawalnagar Campus

Examiner Name: Sohail Murtaza /

Signatures

Candidate Signature

Examiner Signature & Stamp

Principal Signature

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Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- Which term in NLP refers to a curve that plots the performance of a model as a function of a certain parameter?

- (a) Discourse curve
- (b) Pragmatics curve
- (c) Learning curve
- (d) Syntax curve

2- What is the main goal of machine learning?

- (a) To program computers to think like humans
- (b) To enable computers to learn from data and improve performance over time
- (c) To automate tasks without using data
- (d) To optimize computer hardware

3- In Python, which library provides an easy-to-use interface for building and training neural networks, and offers a high-level API for defining layers and model architectures?

- (a) Keras
- (b) TensorFlow
- (c) Scikit-learn
- (d) PyTorch

4- What is the primary purpose of string operations in programming?

- (a) To generate random numbers
- (b) To control robotic movements
- (c) To create complex mathematical equations
- (d) To manipulate and process text

5- What is inheritance in object-oriented programming?

- (a) It allows a class to inherit properties and methods from another class
- (b) It's a way to create new classes from scratch
- (c) It's a way to define unrelated classes
- (d) It's a process of copying and pasting code from one class to another

6- In machine learning, what type of problem involves predicting a continuous numerical output?

- (a) Regression
- (b) Classification
- (c) Reinforcement learning
- (d) Clustering

7- When developing an application that requires understanding and processing natural language input, which Azure service provides a platform for building and managing language understanding models, enabling intent recognition and entity extraction?

- (a) Azure Language Understanding
- (b) Azure Content Moderator
- (c) Azure Text Analytics
- (d) Azure Translator

8- What is the advantage of using functions in programming?

- (a) They allow you to define new data types
- (b) They help you optimize memory usage
- (c) They allow you to create GUI interfaces
- (d) They allow you to reuse code and improve code organization

9- What is the purpose of selecting the appropriate Microsoft Azure AI service for a solution?

- (a) To simplify the code development process
- (b) To ensure the responsible use of AI technologies
- (c) To maximize profits from AI solutions
- (d) To showcase various AI services to clients

10- In Pandas, which method is used to save a DataFrame to a CSV file?

- (a) df.write_csv()
- (b) df.export_csv()
- (c) df.to_csv()
- (d) df.save_csv()

11- In data analysis, what process involves converting continuous numerical data into discrete intervals or bins?

- (a) Binning
- (b) Discretization
- (c) Categorization
- (d) Segmentation

12- When creating an inclusive and accessible application that helps users read content more easily, which Azure service provides features like text-to-speech, translation, and focus mode, enhancing readability for users with varying abilities?

- (a) Azure Personalizer
- (b) Azure Content Moderator
- (c) Azure Text Analytics
- (d) Azure Immersive Reader

13- In neural networks, what are the two main steps involved in the training process, where the first step involves computing the output of the network given the input data, and the second step involves updating the network's weights based on the error?

- (a) Data Preprocessing and Feature Selection
- (b) Forward and Backward Passes
- (c) Hyperparameter Tuning and Cross-Validation
- (d) Model Compilation and Evaluation

14- In supervised machine learning, what is the primary task of the model?

- (a) To generate new data samples
- (b) To preprocess the data
- (c) To cluster similar data points
- (d) To learn from labeled training data to make predictions on new, unseen data

15- In Microsoft Azure, which service provides a platform for building, training, and deploying machine learning models, enabling you to create AI-powered solutions using a variety of tools and frameworks?

- (a) Azure Machine Learning
- (b) Azure AI Builder
- (c) Azure Data Factory
- (d) Azure Cognitive Services

16- Which type of loop is suitable when the number of iterations is known beforehand?

- (a) While loop
- (b) Do-while loop
- (c) For loop
- (d) Infinite loop

17- Which data structure is used to store an unordered collection of unique elements in Python?

- (a) Arrays
- (b) Sets
- (c) Lists
- (d) Tuples

18- What is the primary application of Convolutional Neural Networks (CNNs)?

- (a) Tabular data analysis
- (b) Text classification
- (c) Image and video analysis
- (d) Speech recognition

19- In deep learning, what type of neural network architecture consists of multiple layers of interconnected neurons, including input, hidden, and output layers, where information flows only in one direction from input to output?

- (a) Recurrent Neural Network (RNN)
- (b) Long Short-Term Memory (LSTM)
- (c) MLP Feed Forward Neural Network
- (d) Convolutional Neural Network (CNN)

20- In machine learning, what term refers to a situation where one class has significantly more instances than other classes in a dataset?

- (a) Overfitting
- (b) Dataset augmentation
- (c) Underfitting
- (d) Dataset imbalance

__The End__



Government of Pakistan
National Vocational & Technical Training Commission (NAVTTTC)

Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



Roll no: 1153542

Name: Khadija Ali

CNIC: 3110457550470

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

Test Center Name: DevCastle Training institute |
Software house Bahawalnagar Campus

Examiner Name: Sohail Murtaza /

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Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- When using a cloud service, what is the URL or network address that serves as the entry point for accessing the service's features and functionalities?

- (a) API Key
- (b) Default Endpoint
- (c) Service Manifest
- (d) Authentication Token

2- In deep neural networks, what problem arises when gradients during backpropagation become very small, causing slow convergence and difficulty in training lower layers?

- (a) Overfitting
- (b) Vanishing gradients
- (c) Exploding gradients
- (d) Underfitting

3- What does polymorphism in OOP refer to?

- (a) Overriding parent class methods
- (b) Allowing operators to work with different data types objects as instances of a common superclass
- (c) Creating complex class hierarchies
- (d) Treating different class objects as instances of a common superclass

4- What is the purpose of list comprehension in Python?

- (a) To create complex nested lists
- (b) To manipulate and filter lists using a concise syntax
- (c) To write long and verbose loops
- (d) To generate random numbers

5- What is an object in object-oriented programming?

- (a) A block of code that is executed once
- (b) A function that defines a class
- (c) An instance of a class with its own properties and methods
- (d) A value that can be changed during runtime

6- In machine learning, what type of problem involves predicting a continuous numerical output?

- (a) Regression
- (b) Classification
- (c) Reinforcement learning
- (d) Clustering

7- Which measures assess the spread or dispersion of data points around the mean?

- (a) Mean and Standard Deviation
- (b) Mean and Median
- (c) Variance and Standard Deviation
- (d) Range and Mode

8- In univariate linear regression with vectorization, what advantage does vectorization offer over using loops for computations?

- (a) Reduced memory usage
- (b) Easier code debugging
- (c) Improved computational efficiency
- (d) Enhanced model interpretability

9- In Python, which function is used to apply a function to all elements in an iterable?

- (a) apply()
- (b) filter()
- (c) iterate()
- (d) map()

10- What does Bayesian probability involve?

- (a) Estimating probabilities using random sampling
- (b) Counting the number of occurrences in a dataset
- (c) Calculating probabilities based on prior knowledge and observations
- (d) Using logical operators to find probabilities

11- What is an inner/nested class in OOP?

- (a) A class defined within another class
- (b) A class that can only be instantiated once
- (c) A class that inherits from another class
- (d) A class with no methods or attributes

12- When managing authentication for a cloud resource, which approach involves configuring and using identity and access management (IAM) policies to control who can access and perform actions on the resource?

- (a) IAM Policies
- (b) OAuth Tokens
- (c) API Keys
- (d) Encryption Keys

13- What is the process of breaking a text into individual words or tokens?

- (a) Lemmatization
- (b) Sentence tokenization
- (c) Word tokenization
- (d) Text normalization

14- Which measure quantifies the strength and direction of a linear relationship between two variables?

- (a) Mean
- (b) Correlation Coefficient
- (c) Standard Deviation
- (d) Variance

15- What is the primary application of Convolutional Neural Networks (CNNs)?

- (a) Tabular data analysis
- (b) Text classification
- (c) Image and video analysis
- (d) Speech recognition

16- Which data structure is commonly used in Pandas to store tabular data?

- (a) Arrays
- (b) Lists
- (c) DataFrames
- (d) Sets

17- In neural networks, what graphical representation visualizes the sequence of mathematical operations performed during forward and backward passes, aiding in understanding and optimization of the learning process?

- (a) Loss Function
- (b) Gradient Descent
- (c) Activation Function
- (d) Computational Graph

18- In natural language processing, which model aims to predict a target word based on its surrounding context words, and is often used for learning dense word representations?

- (a) Continuous Bag-of-Words (CBOW)
- (b) Named Entity Recognition
- (c) Part-of-Speech Tagging
- (d) Skip-gram

19- In Linux, which command is used to update the package list?

- (a) apt-get
- (b) apt-upgrade
- (c) apt-list
- (d) apt-update

20- In Pandas, which method is used to save a DataFrame to a CSV file?

- (a) df.write_csv()
- (b) df.export_csv()
- (c) df.to_csv()
- (d) df.save_csv()

__The End__



Government of Pakistan
National Vocational & Technical Training Commission (NAVTTTC)

Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



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CNIC: 3530102938962

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

Test Center Name: DevCastle Training institute |
Software house Bahawalnagar Campus

Examiner Name: Sohail Murtaza /

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- 3) Mobile Phone not allowed.
- 4) Use separate paper for rough work.

Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- Which Seaborn function is used to create a heatmap to visualize the correlation matrix of a DataFrame?

- (a) sns.corrplot()
- (b) sns.matrixplot()
- (c) sns.correlationplot()
- (d) sns.heatmap()

2- In univariate linear regression without vectorization, how is the gradient of the cost function calculated for the slope parameter?

- (a) Using matrix multiplication
- (b) Using a direct formula
- (c) Using a loop to iterate over each data point
- (d) By taking the derivative of the cost function

3- What is the purpose of software installation tools like Anaconda and VSCode in AI development?

- (a) To demonstrate origami techniques
- (b) To install and manage necessary programming environments
- (c) To showcase gardening skills
- (d) To create virtual reality experiences

4- What is the main goal of using exception handling in programming?

- (a) To make code shorter
- (b) To gracefully handle errors and exceptions
- (c) To make code run faster
- (d) To prevent any errors from occurring

5- In univariate linear regression with vectorization, what advantage does vectorization offer over using loops for computations?

- (a) Reduced memory usage
- (b) Easier code debugging
- (c) Improved computational efficiency
- (d) Enhanced model interpretability

6- When developing an application that involves extracting insights, sentiment, and key phrases from text data, which Azure service provides text analytics capabilities for language detection, sentiment analysis, and named entity recognition?

- (a) Azure Content Moderator
- (b) Azure Speech Service
- (c) Azure Text Analytics
- (d) Azure Translator

7- What type of data consists of numerical values that represent quantities?

- (a) Binary data
- (b) Categorical data
- (c) Qualitative data
- (d) Quantitative data

8- In languages without explicit spaces between words, what is the process of dividing a sequence of characters into individual words called?

- (a) Sentence tokenization
- (b) Part-of-speech tagging
- (c) Text normalization
- (d) Word segmentation

9- Which symbol is used in a regular expression to match any single character?

- (a) ^
- (b) *
- (c) .
- (d) ?

10- In programming, what is the term for a function defined inside another function?

- (a) Nested function
- (b) Child function
- (c) Inner function
- (d) Subfunction

11- In machine learning, which loss function is commonly used for classification tasks and measures the dissimilarity between predicted probabilities and actual class labels?

- (a) Mean Absolute Error
- (b) Mean Squared Error
- (c) Huber Loss
- (d) Cross-Entropy

12- In natural language processing, what is the process of grouping adjacent words together into meaningful phrases or "chunks" based on their part-of-speech tags?

- (a) Chunking
- (b) NER tagging
- (c) Chinking
- (d) POS tagging

13- What is the primary application of Convolutional Neural Networks (CNNs)?

- (a) Tabular data analysis
- (b) Text classification
- (c) Image and video analysis
- (d) Speech recognition

14- Seaborn is a Python data visualization library based on which other library?

- (a) Scikit-learn
- (b) Matplotlib
- (c) Pandas
- (d) Numpy

15- Which measure provides an indication of how spread out the values in a dataset are?

- (a) Z-Score
- (b) Coefficient of Variation
- (c) Variance
- (d) Standard Deviation

16- In statistics, which term describes a variable that can take on distinct, separate values?

- (a) Quantitative
- (b) Qualitative
- (c) Discrete
- (d) Continuous

17- In machine learning, which algorithm creates a tree-like model of decisions and their possible consequences, where each internal node represents a decision based on an attribute, and each leaf node represents an outcome?

- (a) Linear Regression
- (b) Decision Tree
- (c) K-Means Clustering
- (d) Support Vector Machine (SVM)

18- Which Pandas plotting method is used to create a horizontal bar plot?

- (a) `df.plot.hist()`
- (b) `df.plot.barh()`
- (c) `df.plot.bar()`
- (d) `df.plot.line()`

19- In Python, which library provides an easy-to-use interface for building and training neural networks, and offers a high-level API for defining layers and model architectures?

- (a) Keras
- (b) TensorFlow
- (c) Scikit-learn
- (d) PyTorch

20- What field of study focuses on enabling computers to understand, interpret, and generate human language?

- (a) Computer Vision
- (b) Data Science
- (c) Natural Language Processing
- (d) Machine Learning

__The End__



Government of Pakistan
National Vocational & Technical Training Commission (NAVTTTC)

Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



Roll no: 1153532

Name: Abdul Rauf Shehzad

CNIC: 3110518783995

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

Test Center Name: DevCastle Training institute |
Software house Bahawalnagar Campus

Examiner Name: Sohail Murtaza /

Signatures

Candidate Signature

Examiner Signature & Stamp

Principal Signature

Instructions:

Please read instructions carefully before attempting paper:

- 1) All questions are compulsory.
- 2) The question paper consists of 21 questions divided into two sections A & B. Section A comprise of 20 MCQs of 1 marks each. Section B comprises of 1 practical task having 80 marks.
- 3) Mobile Phone not allowed.
- 4) Use separate paper for rough work.

Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- When managing authentication for a cloud resource, which approach involves configuring and using identity and access management (IAM) policies to control who can access and perform actions on the resource?

- (a) IAM Policies
- (b) OAuth Tokens
- (c) API Keys
- (d) Encryption Keys

2- Which type of sequence model is used for tasks like machine translation?

- (a) Feedforward Neural Networks
- (b) Recurrent Neural Networks (RNN)
- (c) Convolutional Neural Networks (CNN)
- (d) Decision Trees

3- What field of study focuses on enabling computers to understand, interpret, and generate human language?

- (a) Computer Vision
- (b) Data Science
- (c) Natural Language Processing
- (d) Machine Learning

4- In statistics, which term describes a variable that can take on distinct, separate values?

- (a) Quantitative
- (b) Qualitative
- (c) Discrete
- (d) Continuous

5- In natural language processing, which model aims to predict a target word based on its surrounding context words, and is often used for learning dense word representations?

- (a) Continuous Bag-of-Words (CBOW)
- (b) Named Entity Recognition
- (c) Part-of-Speech Tagging
- (d) Skip-gram

6- When integrating Azure AI services into a continuous integration/continuous deployment (CI/CD) pipeline, which practice involves automating the deployment of AI models and services to production environments, ensuring consistent and efficient deployment p

- (a) CI/CD Integration
- (b) Data Collection
- (c) Model Training
- (d) Feature Engineering

7- Which keyword is used in Python to introduce an alternative condition if the initial condition is false?

- (a) Elself
- (b) Otherwise
- (c) Else
- (d) Unless

8- In machine learning, which algorithm creates a tree-like model of decisions and their possible consequences, where each internal node represents a decision based on an attribute, and each leaf node represents an outcome?

- (a) Linear Regression
- (b) Decision Tree
- (c) K-Means Clustering
- (d) Support Vector Machine (SVM)

9- In NLP, what is the process of reducing words to their base or dictionary form while ensuring they are still valid words?

- (a) Lemmatization
- (b) Tokenization
- (c) Chunking
- (d) Stemming

10- What is the process of breaking a text into individual words or tokens?

- (a) Lemmatization
- (b) Sentence tokenization
- (c) Word tokenization
- (d) Text normalization

11- In Python, what is the purpose of file handling?

- (a) To delete unnecessary files
- (b) To control the display of files in the terminal
- (c) To create backups of important files
- (d) To read from and write to files on disk

12- Which measure provides information about the relative position of a data point within a dataset?

- (a) Mean
- (b) Standard Deviation
- (c) Percentile
- (d) Range

13- In Pandas, which data structure is used to store one-dimensional labeled data, similar to a column in a spreadsheet?

- (a) List
- (b) DataFrame
- (c) Array
- (d) Series

14- What are universal array functions in Numpy?

- (a) Functions that perform matrix multiplication
- (b) Functions that operate element-wise on arrays
- (c) Functions that only work on 1D arrays
- (d) Functions that require external libraries

15- What is the purpose of selecting the appropriate Microsoft Azure AI service for a solution?

- (a) To simplify the code development process
- (b) To ensure the responsible use of AI technologies
- (c) To maximize profits from AI solutions
- (d) To showcase various AI services to clients

16- In univariate linear regression with gradient descent, what is the role of the gradient descent algorithm?

- (a) To calculate the slope of the regression line
- (b) To maximize the cost function and find optimal parameters
- (c) To minimize the cost function and find optimal parameters
- (d) To calculate the intercept of the regression line

17- What type of data consists of numerical values that represent quantities?

- (a) Binary data
- (b) Categorical data
- (c) Qualitative data
- (d) Quantitative data

18- In supervised machine learning, which component represents the function that the model is trying to learn and approximate?

- (a) Optimizer
- (b) Hypothesis
- (c) Cost function
- (d) Labeled data

19- Seaborn is a Python data visualization library based on which other library?

- (a) Scikit-learn
- (b) Matplotlib
- (c) Pandas
- (d) Numpy

20- What does multiple inheritance in OOP involve?

- (a) Inheriting from a grandparent class
- (b) Inheriting from a child class
- (c) Inheriting from a single parent class
- (d) Inheriting from multiple parent classes

__The End__



Government of Pakistan
National Vocational & Technical Training Commission (NAVTTTC)

Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



Roll no: 1153537

Name: Awais Qarni

CNIC: 3110554071975

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

Test Center Name: DevCastle Training institute |
Software house Bahawalnagar Campus

Examiner Name: Sohail Murtaza /

Signatures

Candidate Signature

Examiner Signature & Stamp

Principal Signature

Instructions:

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- 3) Mobile Phone not allowed.
- 4) Use separate paper for rough work.

Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- What is the process of breaking a text into individual words or tokens?

- (a) Lemmatization
- (b) Sentence tokenization
- (c) Word tokenization
- (d) Text normalization

2- In machine learning, what is the process of selecting a subset of relevant features from the original set of features to improve model performance and reduce complexity?

- (a) Feature extraction
- (b) Feature selection
- (c) NER tagging
- (d) Tokenization

3- What is dynamic polymorphism in object-oriented programming?

- (a) A technique to combine multiple classes into one
- (b) The ability of a subclass to override methods of its base class
- (c) A way to prevent inheritance between classes
- (d) The process of defining new data types

4- In Microsoft Azure, what networking solution allows you to isolate and secure your cloud resources by creating private network connections and controlling traffic flow between virtual machines and services?

- (a) Azure Active Directory
- (b) Azure Firewall
- (c) Azure Virtual Networks
- (d) Azure Security Center

5- In deep learning, which type of neural network is commonly used for image classification tasks, leveraging 2D convolutional layers to automatically learn hierarchical features from images?

- (a) 2D Convolutional Neural Network (CNN)
- (b) Autoencoder
- (c) Multi-Layer Perceptron (MLP)
- (d) Recurrent Neural Network (RNN)

6- When integrating Azure AI services into a continuous integration/continuous deployment (CI/CD) pipeline, which practice involves automating the deployment of AI models and services to production environments, ensuring consistent and efficient deployment p

- (a) CI/CD Integration
- (b) Data Collection
- (c) Model Training
- (d) Feature Engineering

7- In univariate linear regression with gradient descent, what is the role of the gradient descent algorithm?

- (a) To calculate the slope of the regression line
- (b) To maximize the cost function and find optimal parameters
- (c) To minimize the cost function and find optimal parameters
- (d) To calculate the intercept of the regression line

8- In natural language processing, which model aims to predict a target word based on its surrounding context words, and is often used for learning dense word representations?

- (a) Continuous Bag-of-Words (CBOW)
- (b) Named Entity Recognition
- (c) Part-of-Speech Tagging
- (d) Skip-gram

9- In Python, which function is used to apply a function to all elements in an iterable?

- (a) apply()
- (b) filter()
- (c) iterate()
- (d) map()

10- Which measure provides information about the relative position of a data point within a dataset?

- (a) Mean
- (b) Standard Deviation
- (c) Percentile
- (d) Range

11- What is the primary purpose of string operations in programming?

- (a) To generate random numbers
- (b) To control robotic movements
- (c) To create complex mathematical equations
- (d) To manipulate and process text

12- Which Numpy function is used to create a copy of an array?

- (a) `numpy.clone()`
- (b) `numpy.copy()`
- (c) `numpy.duplicate()`
- (d) `numpy.replicate()`

13- Which linguistic component focuses on the structure and rules governing the arrangement of words and phrases to form sentences?

- (a) Discourse
- (b) Semantics
- (c) Syntax
- (d) Pragmatics

14- In machine learning, what term refers to a situation where one class has significantly more instances than other classes in a dataset?

- (a) Overfitting
- (b) Dataset augmentation
- (c) Underfitting
- (d) Dataset imbalance

15- What is the primary purpose of using nested loops in programming?

- (a) To reduce code complexity
- (b) To create recursive functions
- (c) To execute a loop inside another loop
- (d) To improve code readability

16- In machine learning, which algorithm is used for both classification and regression tasks and aims to find the hyperplane that best separates data points of different classes while maximizing the margin?

- (a) Linear Regression
- (b) Support Vector Machine (SVM)
- (c) Decision Tree
- (d) K-Means Clustering

17- When planning for an AI solution that aligns with Responsible AI principles, which practice involves ensuring transparency and explainability of AI models and decisions, enabling users to understand the reasoning behind predictions?

- (a) Model Accuracy
- (b) Explainability and Transparency
- (c) Performance Metrics
- (d) Data Collection

18- Which data structure is used to store an unordered collection of unique elements in Python?

- (a) Arrays
- (b) Sets
- (c) Lists
- (d) Tuples

19- Which type of loop is suitable when the number of iterations is known beforehand?

- (a) While loop
- (b) Do-while loop
- (c) For loop
- (d) Infinite loop

20- Which measure provides an indication of how spread out the values in a dataset are?

- (a) Z-Score
- (b) Coefficient of Variation
- (c) Variance
- (d) Standard Deviation

__The End__



Government of Pakistan
National Vocational & Technical Training Commission (NAVTTTC)

Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



Roll no: 1153540

Name: Aysha nadeem

CNIC: 3110556331864

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

Test Center Name: DevCastle Training institute |
Software house Bahawalnagar Campus

Examiner Name: Sohail Murtaza /

Signatures

Candidate Signature

Examiner Signature & Stamp

Principal Signature

Instructions:

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- 3) Mobile Phone not allowed.
- 4) Use separate paper for rough work.

Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- In univariate linear regression with gradient descent, what is the role of the gradient descent algorithm?

- (a) To calculate the slope of the regression line
- (b) To maximize the cost function and find optimal parameters
- (c) To minimize the cost function and find optimal parameters
- (d) To calculate the intercept of the regression line

2- In machine learning, what term refers to a situation where a model learns the training data too well, capturing noise and leading to poor generalization on new data?

- (a) Bias-Variance Tradeoff
- (b) Regularization
- (c) Underfitting
- (d) Overfitting

3- What is a lexical database in NLP that provides a semantic network of words and their relationships, including synonyms and hypernyms?

- (a) NER tagging
- (b) Stemming
- (c) POS tagging
- (d) WordNet

4- In statistics, which term describes a variable that can take on distinct, separate values?

- (a) Quantitative
- (b) Qualitative
- (c) Discrete
- (d) Continuous

5- What is the advantage of using functions in programming?

- (a) They allow you to define new data types
- (b) They help you optimize memory usage
- (c) They allow you to create GUI interfaces
- (d) They allow you to reuse code and improve code organization

6- What field of study focuses on enabling computers to understand, interpret, and generate human language?

- (a) Computer Vision
- (b) Data Science
- (c) Natural Language Processing
- (d) Machine Learning

7- What is hierarchical inheritance in OOP?

- (a) Inheriting from a single parent class
- (b) Inheriting from a grandparent class
- (c) Inheriting from multiple child classes
- (d) Inheriting from multiple parent classes

8- What is an inner/nested class in OOP?

- (a) A class defined within another class
- (b) A class that can only be instantiated once
- (c) A class that inherits from another class
- (d) A class with no methods or attributes

9- In logistic regression for multiclass classification, what type of decision boundary is typically learned by the model?

- (a) Exponential decision boundary
- (b) Linear decision boundary
- (c) Non-linear decision boundary
- (d) Constant decision boundary

10- What does multiple inheritance in OOP involve?

- (a) Inheriting from a grandparent class
- (b) Inheriting from a child class
- (c) Inheriting from a single parent class
- (d) Inheriting from multiple parent classes

11- What is the process of breaking a text into individual words or tokens?

- (a) Lemmatization
- (b) Sentence tokenization
- (c) Word tokenization
- (d) Text normalization

12- Which technique in natural language processing is used to convert words into dense vector representations, capturing semantic meanings and relationships between words?

- (a) Word2Vec
- (b) Named Entity Recognition
- (c) Bag-of-Words
- (d) POS tagging

13- In machine learning, which ensemble method combines multiple decision trees by training them on different subsets of the training data and averaging their predictions to improve model accuracy and reduce overfitting?

- (a) Random Forest
- (b) K-Means Clustering
- (c) Gradient Boosting
- (d) AdaBoost

14- In neural networks, what are the two main steps involved in the training process, where the first step involves computing the output of the network given the input data, and the second step involves updating the network's weights based on the error?

- (a) Data Preprocessing and Feature Selection
- (b) Forward and Backward Passes
- (c) Hyperparameter Tuning and Cross-Validation
- (d) Model Compilation and Evaluation

15- In programming, what is the purpose of a function?

- (a) To organize code into reusable blocks
- (b) To create loops
- (c) To print output
- (d) To store data

16- In machine learning, what term refers to a situation where one class has significantly more instances than other classes in a dataset?

- (a) Overfitting
- (b) Dataset augmentation
- (c) Underfitting
- (d) Dataset imbalance

17- What is type casting in Numpy used for?

- (a) Changing the shape of an array
- (b) Changing the data type of an array
- (c) Changing the dimensions of an array
- (d) Changing the elements of an array

18- What does polymorphism in OOP refer to?

- (a) Overriding parent class methods
- (b) Allowing operators to work with different data types objects as instances of a common superclass
- (c) Creating complex class hierarchies
- (d) Treating different class objects as instances of a common superclass

19- When deploying a machine learning model as a web service in Microsoft Azure, which service allows you to create, deploy, and manage RESTful APIs for your trained models, making them accessible for predictions?

- (a) Azure Cognitive Services
- (b) Azure Functions
- (c) Azure Logic Apps
- (d) Azure Machine Learning Service

20- If you're building a speech solution that involves converting spoken language into text, performing text-to-speech synthesis, and enabling speaker recognition, which Azure Cognitive Services offering is best suited for these tasks?

- (a) Azure Text Analytics
- (b) Azure Language Understanding
- (c) Azure Speech Service
- (d) Azure Translator

__The End__



Government of Pakistan
National Vocational & Technical Training Commission (NAVTTTC)

Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



Roll no: 1153536

Name: Javeria Tabassum

CNIC: 3110165059134

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

Test Center Name: DevCastle Training institute |
Software house Bahawalnagar Campus

Examiner Name: Sohail Murtaza /

Signatures

Candidate Signature

Examiner Signature & Stamp

Principal Signature

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- 3) Mobile Phone not allowed.
- 4) Use separate paper for rough work.

Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- Which command is used to change the permissions of a file in Unix-like systems?

- (a) touch
- (b) kill
- (c) chmod
- (d) ps

2- What does Bayesian probability involve?

- (a) Estimating probabilities using random sampling
- (b) Counting the number of occurrences in a dataset
- (c) Calculating probabilities based on prior knowledge and observations
- (d) Using logical operators to find probabilities

3- What does the term "probability" refer to in statistics?

- (a) The ratio of successful outcomes to the total possible outcomes
- (b) The number of trials in an experiment
- (c) The certainty of an event occurring
- (d) The number of times an event occurs

4- Which command is used to display the contents of a file in a Unix-like operating system?

- (a) read
- (b) view
- (c) cat
- (d) open

5- When developing a solution that automatically detects and filters inappropriate or offensive content from text, images, and videos, which Azure service provides the capability to moderate and classify content to ensure it meets appropriate standards?

- (a) Azure Cognitive Services
- (b) Azure Content Moderator
- (c) Azure Machine Learning
- (d) Azure Computer Vision

6- In the context of data, what does "structured" data refer to?

- (a) Data that is randomly generated
- (b) Data that is not organized in any specific way
- (c) Data that is not useful for analysis
- (d) Data that is organized into a specific format or schema

7- Which Seaborn function is used to create a bar plot of categorical data?

- (a) sns.catplot()
- (b) sns.barplot()
- (c) sns.scatterplot()
- (d) sns.lineplot()

8- What is the advantage of using functions in programming?

- (a) They allow you to define new data types
- (b) They help you optimize memory usage
- (c) They allow you to create GUI interfaces
- (d) They allow you to reuse code and improve code organization

9- In Microsoft Azure, what networking solution allows you to isolate and secure your cloud resources by creating private network connections and controlling traffic flow between virtual machines and services?

- (a) Azure Active Directory
- (b) Azure Firewall
- (c) Azure Virtual Networks
- (d) Azure Security Center

10- Which Numpy function is used to create a copy of an array?

- (a) numpy.clone()
- (b) numpy.copy()
- (c) numpy.duplicate()
- (d) numpy.replicate()

11- Which term in NLP refers to a curve that plots the performance of a model as a function of a certain parameter?

- (a) Discourse curve
- (b) Pragmatics curve
- (c) Learning curve
- (d) Syntax curve

12- What is the purpose of "magic functions" or "dunder methods" in Python?

- (a) Enforcing strict naming conventions
- (b) Defining private methods in a class
- (c) Creating unique variable names
- (d) Overloading operators and providing special behavior

13- In univariate linear regression with vectorization, what advantage does vectorization offer over using loops for computations?

- (a) Reduced memory usage
- (b) Easier code debugging
- (c) Improved computational efficiency
- (d) Enhanced model interpretability

14- In deep learning, what type of neural network architecture consists of multiple layers of interconnected neurons, including input, hidden, and output layers, where information flows only in one direction from input to output?

- (a) Recurrent Neural Network (RNN)
- (b) Long Short-Term Memory (LSTM)
- (c) MLP Feed Forward Neural Network
- (d) Convolutional Neural Network (CNN)

15- What is the primary purpose of using nested loops in programming?

- (a) To reduce code complexity
- (b) To create recursive functions
- (c) To execute a loop inside another loop
- (d) To improve code readability

16- In object-oriented programming, what is an abstract method?

- (a) A method that can't be called
- (b) A method that is automatically generated
- (c) A method that has no implementation
- (d) A method that can't be overridden

17- What is an inner/nested class in OOP?

- (a) A class defined within another class
- (b) A class that can only be instantiated once
- (c) A class that inherits from another class
- (d) A class with no methods or attributes

18- When developing an application that requires translating text from one language to another, which Azure service provides language translation capabilities and supports a wide range of languages for translation?

- (a) Azure Text Analytics
- (b) Azure Content Moderator
- (c) Azure Speech Service
- (d) Azure Translator

19- In the classical machine learning pipeline, what step involves transforming raw data into a suitable format for analysis?

- (a) Feature crafting
- (b) Data preprocessing
- (c) Modeling
- (d) Testing and validation

20- Which evaluation metric is commonly used for binary classification problems and calculates the ratio of correctly predicted positive observations to the total predicted positives?

- (a) Accuracy
- (b) Precision
- (c) F1-score
- (d) Recall

__The End__



Government of Pakistan
National Vocational & Technical Training Commission (NAVTTTC)

Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



Roll no: 1153535

Name: Sohail Asghar

CNIC: 3110122960941

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

Test Center Name: DevCastle Training institute |
Software house Bahawalnagar Campus

Examiner Name: Sohail Murtaza /

Signatures

Candidate Signature

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- 3) Mobile Phone not allowed.
- 4) Use separate paper for rough work.

Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- What is the primary purpose of string operations in programming?

- (a) To generate random numbers
- (b) To control robotic movements
- (c) To create complex mathematical equations
- (d) To manipulate and process text

2- What is an object in object-oriented programming?

- (a) A block of code that is executed once
- (b) A function that defines a class
- (c) An instance of a class with its own properties and methods
- (d) A value that can be changed during runtime

3- In Python, which library provides an easy-to-use interface for building and training neural networks, and offers a high-level API for defining layers and model architectures?

- (a) Keras
- (b) TensorFlow
- (c) Scikit-learn
- (d) PyTorch

4- Which measure quantifies the relative variability of a dataset, considering the standard deviation and mean?

- (a) Coefficient of Variation
- (b) Skewness
- (c) Kurtosis
- (d) Variance

5- In machine learning, what term refers to a situation where one class has significantly more instances than other classes in a dataset?

- (a) Overfitting
- (b) Dataset augmentation
- (c) Underfitting
- (d) Dataset imbalance

6- Which command is used to display the contents of a directory in the terminal?

- (a) ls
- (b) display
- (c) dir
- (d) show

7- What is the purpose of list comprehension in Python?

- (a) To create complex nested lists
- (b) To manipulate and filter lists using a concise syntax
- (c) To write long and verbose loops
- (d) To generate random numbers

8- Which Numpy function is used to create a copy of an array?

- (a) `numpy.clone()`
- (b) `numpy.copy()`
- (c) `numpy.duplicate()`
- (d) `numpy.replicate()`

9- Which measure of central tendency is most affected by outliers?

- (a) Standard Deviation
- (b) Mean
- (c) Median
- (d) Mode

10- In probability theory, what type of probability considers the likelihood of events occurring together?

- (a) Joint probability
- (b) Conditional probability
- (c) Independent probability
- (d) Marginal probability

11- Which data structure is commonly used in Pandas to store tabular data?

- (a) Arrays
- (b) Lists
- (c) DataFrames
- (d) Sets

12- In NLP, what is the process of converting text to a consistent format, such as converting all characters to lowercase and removing special characters?

- (a) Stemming
- (b) Part-of-speech tagging
- (c) Tokenization
- (d) Text normalization

13- When building a solution that involves converting spoken language into written text and performing speech-to-text transcription, which Azure service offers automatic speech recognition capabilities for converting spoken language into text data?

- (a) Azure Translator
- (b) Azure Text Analytics
- (c) Azure Speech Service
- (d) Azure Content Moderator

14- In univariate linear regression with vectorization, what advantage does vectorization offer over using loops for computations?

- (a) Reduced memory usage
- (b) Easier code debugging
- (c) Improved computational efficiency
- (d) Enhanced model interpretability

15- What are the fundamental building blocks of programming that manipulate data?

- (a) Operators
- (b) Keywords
- (c) Variables
- (d) Functions

16- What does polymorphism in OOP refer to?

- (a) Overriding parent class methods
- (b) Allowing operators to work with different data types objects as instances of a common superclass
- (c) Creating complex class hierarchies
- (d) Treating different class objects as instances of a common superclass

17- When planning for an AI solution that aligns with Responsible AI principles, which practice involves ensuring transparency and explainability of AI models and decisions, enabling users to understand the reasoning behind predictions?

- (a) Model Accuracy
- (b) Explainability and Transparency
- (c) Performance Metrics
- (d) Data Collection

18- If you're developing an AI solution that involves extracting meaningful insights from unstructured text data, identifying sentiment, entities, and key phrases, which cloud service offers a comprehensive Text Analytics API?

- (a) Azure Cognitive Services
- (b) IBM Watson
- (c) Google Cloud AI
- (d) Amazon SageMaker

19- In the classical machine learning pipeline, what step involves transforming raw data into a suitable format for analysis?

- (a) Feature crafting
- (b) Data preprocessing
- (c) Modeling
- (d) Testing and validation

20- In machine learning, which task involves predicting discrete class labels as output?

- (a) Dimensionality reduction
- (b) Classification
- (c) Regression
- (d) Clustering

__The End__



Government of Pakistan
National Vocational & Technical Training Commission (NAVTTTC)

Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



Roll no: 1153534

Name: Ahsaan azam

CNIC: 3640159206199

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

Test Center Name: DevCastle Training institute |
Software house Bahawalnagar Campus

Examiner Name: Sohail Murtaza /

Signatures

Candidate Signature

Examiner Signature & Stamp

Principal Signature

Instructions:

Please read instructions carefully before attempting paper:

- 1) All questions are compulsory.
- 2) The question paper consists of 21 questions divided into two sections A & B. Section A comprise of 20 MCQs of 1 marks each. Section B comprises of 1 practical task having 80 marks.
- 3) Mobile Phone not allowed.
- 4) Use separate paper for rough work.

Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- What is inheritance in object-oriented programming?

- (a) It allows a class to inherit properties and methods from another class
- (b) It's a way to create new classes from scratch
- (c) It's a way to define unrelated classes
- (d) It's a process of copying and pasting code from one class to another

2- In Microsoft Azure, what networking solution allows you to isolate and secure your cloud resources by creating private network connections and controlling traffic flow between virtual machines and services?

- (a) Azure Active Directory
- (b) Azure Firewall
- (c) Azure Virtual Networks
- (d) Azure Security Center

3- When integrating Azure AI services into a continuous integration/continuous deployment (CI/CD) pipeline, which practice involves automating the deployment of AI models and services to production environments, ensuring consistent and efficient deployment p

- (a) CI/CD Integration
- (b) Data Collection
- (c) Model Training
- (d) Feature Engineering

4- In machine learning, what term refers to a situation where one class has significantly more instances than other classes in a dataset?

- (a) Overfitting
- (b) Dataset augmentation
- (c) Underfitting
- (d) Dataset imbalance

5- What is the purpose of software installation tools like Anaconda and VSCode in AI development?

- (a) To demonstrate origami techniques
- (b) To install and manage necessary programming environments
- (c) To showcase gardening skills
- (d) To create virtual reality experiences

6- If you're developing an AI solution that involves extracting meaningful insights from unstructured text data, identifying sentiment, entities, and key phrases, which cloud service offers a comprehensive Text Analytics API?

- (a) Azure Cognitive Services
- (b) IBM Watson
- (c) Google Cloud AI
- (d) Amazon SageMaker

7- Which measure indicates how many standard deviations a data point is away from the mean?

- (a) Quartile
- (b) Z-Score
- (c) Range
- (d) Percentile

8- What is type casting in Numpy used for?

- (a) Changing the shape of an array
- (b) Changing the data type of an array
- (c) Changing the dimensions of an array
- (d) Changing the elements of an array

9- What type of data consists of numerical values that represent quantities?

- (a) Binary data
- (b) Categorical data
- (c) Qualitative data
- (d) Quantitative data

10- Which type of data includes non-numerical information such as labels or categories?

- (a) Numerical data
- (b) Qualitative data
- (c) Categorical data
- (d) Quantitative data

11- Which technique in natural language processing is used to convert words into dense vector representations, capturing semantic meanings and relationships between words?

- (a) Word2Vec
- (b) Named Entity Recognition
- (c) Bag-of-Words
- (d) POS tagging

12- In natural language processing, what is the process of reducing a word to its base or root form called, by removing suffixes or prefixes?

- (a) Segmentation
- (b) Stemming
- (c) Lemmatization
- (d) Tokenization

13- Which linguistic component focuses on the structure and rules governing the arrangement of words and phrases to form sentences?

- (a) Discourse
- (b) Semantics
- (c) Syntax
- (d) Pragmatics

14- In machine learning, what type of problem involves predicting a continuous numerical output?

- (a) Regression
- (b) Classification
- (c) Reinforcement learning
- (d) Clustering

15- What method is used to select elements from a Numpy array based on a condition?

- (a) conditional_selection()
- (b) filter()
- (c) logical_select()
- (d) where()

16- Which Numpy function is used to create a copy of an array?

- (a) numpy.clone()
- (b) numpy.copy()
- (c) numpy.duplicate()
- (d) numpy.replicate()

17- Which type of plot displays data for a single variable, showing its distribution and characteristics?

- (a) Bivariate plot
- (b) Multivariate plot
- (c) Scatter plot
- (d) Univariate plot

18- In cloud services, what are used as credentials to securely authenticate and access resources, and are often required to be protected and rotated regularly to enhance security?

- (a) API keys
- (b) Account keys
- (c) Access tokens
- (d) Encryption keys

19- How does the Anomaly Detector service in Microsoft Azure work?

- (a) It identifies abnormalities based on statistical analysis
- (b) It compares data to pre-defined patterns
- (c) It relies on user-defined rules to detect anomalies
- (d) It uses machine learning to predict anomalies

20- In Pandas, which data structure is used to store one-dimensional labeled data, similar to a column in a spreadsheet?

- (a) List
- (b) DataFrame
- (c) Array
- (d) Series

__The End__



Government of Pakistan
National Vocational & Technical Training Commission (NAVTTTC)

Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



Roll no: 1153533

Name: Mudassar Hussain

CNIC: 3110152368247

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

Test Center Name: DevCastle Training institute |
Software house Bahawalnagar Campus

Examiner Name: Sohail Murtaza /

Signatures

Candidate Signature

Examiner Signature & Stamp

Principal Signature

Instructions:

Please read instructions carefully before attempting paper:

- 1) All questions are compulsory.
- 2) The question paper consists of 21 questions divided into two sections A & B. Section A comprise of 20 MCQs of 1 marks each. Section B comprises of 1 practical task having 80 marks.
- 3) Mobile Phone not allowed.
- 4) Use separate paper for rough work.

Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- What is the primary purpose of string operations in programming?

- (a) To generate random numbers
- (b) To control robotic movements
- (c) To create complex mathematical equations
- (d) To manipulate and process text

2- What is the advantage of using functions in programming?

- (a) They allow you to define new data types
- (b) They help you optimize memory usage
- (c) They allow you to create GUI interfaces
- (d) They allow you to reuse code and improve code organization

3- In machine learning, which ensemble method combines multiple decision trees by training them on different subsets of the training data and averaging their predictions to improve model accuracy and reduce overfitting?

- (a) Random Forest
- (b) K-Means Clustering
- (c) Gradient Boosting
- (d) AdaBoost

4- In object-oriented programming, what is multilevel inheritance?

- (a) Inheriting from multiple parent classes
- (b) Inheriting from a child class
- (c) Inheriting from a grandparent class
- (d) Inheriting from a single parent class

5- What type of data consists of numerical values that represent quantities?

- (a) Binary data
- (b) Categorical data
- (c) Qualitative data
- (d) Quantitative data

6- In data analysis, what process involves converting continuous numerical data into discrete intervals or bins?

- (a) Binning
- (b) Discretization
- (c) Categorization
- (d) Segmentation

7- In univariate linear regression without vectorization, how is the gradient of the cost function calculated for the slope parameter?

- (a) Using matrix multiplication
- (b) Using a direct formula
- (c) Using a loop to iterate over each data point
- (d) By taking the derivative of the cost function

8- In probability theory, what type of probability considers the likelihood of events occurring together?

- (a) Joint probability
- (b) Conditional probability
- (c) Independent probability
- (d) Marginal probability

9- Which measure quantifies the strength and direction of a linear relationship between two variables?

- (a) Mean
- (b) Correlation Coefficient
- (c) Standard Deviation
- (d) Variance

10- In natural language processing, what is the process of grouping adjacent words together into meaningful phrases or "chunks" based on their part-of-speech tags?

- (a) Chunking
- (b) NER tagging
- (c) Chinking
- (d) POS tagging

11- Which service can you use to analyze images and extract text from them in Microsoft Azure?

- (a) Anomaly Detector
- (b) Personalizer
- (c) Metrics Advisor
- (d) Content Moderator

12- How can you create a Numpy array from a Python list?

- (a) `numpy.create()`
- (b) `numpy.array()`
- (c) `numpy.from_list()`
- (d) `numpy.list()`

13- In neural networks, what are the two main steps involved in the training process, where the first step involves computing the output of the network given the input data, and the second step involves updating the network's weights based on the error?

- (a) Data Preprocessing and Feature Selection
- (b) Forward and Backward Passes
- (c) Hyperparameter Tuning and Cross-Validation
- (d) Model Compilation and Evaluation

14- Which measure of central tendency is most affected by outliers?

- (a) Standard Deviation
- (b) Mean
- (c) Median
- (d) Mode

15- In programming, what is the purpose of a function?

- (a) To organize code into reusable blocks
- (b) To create loops
- (c) To print output
- (d) To store data

16- What is the primary application of natural language processing (NLP) solutions?

- (a) Image analysis
- (b) Video compression
- (c) Text analysis and understanding
- (d) Audio processing

17- In machine learning, which loss function is commonly used for classification tasks and measures the dissimilarity between predicted probabilities and actual class labels?

- (a) Mean Absolute Error
- (b) Mean Squared Error
- (c) Huber Loss
- (d) Cross-Entropy

18- In statistics, which term describes a variable that can take on distinct, separate values?

- (a) Quantitative
- (b) Qualitative
- (c) Discrete
- (d) Continuous

19- Which Seaborn function is used to create a heatmap to visualize the correlation matrix of a DataFrame?

- (a) `sns.corrplot()`
- (b) `sns.matrixplot()`
- (c) `sns.correlationplot()`
- (d) `sns.heatmap()`

20- In the Bag-of-Words (BoW) model, how is a text represented as a feature vector for NLP tasks?

- (a) By using lemmatization
- (b) By counting the frequency of words in the text
- (c) By removing stopwords
- (d) By converting words to lowercase

__The End__



Government of Pakistan
National Vocational & Technical Training Commission (NAVTTTC)

Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



Roll no: 1153538

Name: Muhammad Muzamil

CNIC: 3110108575687

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

Test Center Name: DevCastle Training institute |
Software house Bahawalnagar Campus

Examiner Name: Sohail Murtaza /

Signatures

Candidate Signature

Examiner Signature & Stamp

Principal Signature

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- 4) Use separate paper for rough work.

Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- When planning for an AI solution that aligns with Responsible AI principles, which practice involves ensuring transparency and explainability of AI models and decisions, enabling users to understand the reasoning behind predictions?

- (a) Model Accuracy
- (b) Explainability and Transparency
- (c) Performance Metrics
- (d) Data Collection

2- How can you convert a user's input into a numerical data type in programming?

- (a) By performing a bitwise operation
- (b) Through the input() function and type casting
- (c) Using the stringify() function
- (d) By using the floatify() method

3- Which symbol is used in a regular expression to match any single character?

- (a) ^
- (b) *
- (c) .
- (d) ?

4- When creating a new resource in Microsoft Azure, which web-based interface provides a graphical user interface (GUI) that allows you to select and configure various Azure services and resources?

- (a) Azure PowerShell
- (b) Azure Portal
- (c) Azure CLI
- (d) Azure SDK

5- What type of data consists of numerical values that represent quantities?

- (a) Binary data
- (b) Categorical data
- (c) Qualitative data
- (d) Quantitative data

6- Which evaluation metric is commonly used for binary classification problems and calculates the ratio of correctly predicted positive observations to the total predicted positives?

- (a) Accuracy
- (b) Precision
- (c) F1-score
- (d) Recall

7- What is the advantage of using functions in programming?

- (a) They allow you to define new data types
- (b) They help you optimize memory usage
- (c) They allow you to create GUI interfaces
- (d) They allow you to reuse code and improve code organization

8- Which measure of central tendency is most affected by outliers?

- (a) Standard Deviation
- (b) Mean
- (c) Median
- (d) Mode

9- In machine learning, which task involves predicting discrete class labels as output?

- (a) Dimensionality reduction
- (b) Classification
- (c) Regression
- (d) Clustering

10- In the classical machine learning pipeline, what step involves transforming raw data into a suitable format for analysis?

- (a) Feature crafting
- (b) Data preprocessing
- (c) Modeling
- (d) Testing and validation

11- In Pandas, what operation allows you to group a DataFrame based on a column or columns and then perform aggregate functions on each group?

- (a) GroupBy
- (b) Aggregate
- (c) Group
- (d) Combine

12- What is the measure of central tendency that represents the average value of a dataset?

- (a) Mode
- (b) Median
- (c) Mean
- (d) Range

13- If you're developing a natural language processing solution that involves sentiment analysis, named entity recognition, and language translation, which cloud service would be a suitable choice for its comprehensive language analysis capabilities?

- (a) Google Cloud Natural Language
- (b) Azure Cognitive Services
- (c) IBM Watson Natural Language Understanding
- (d) Amazon Comprehend

14- In natural language processing, what is the process of assigning grammatical tags to words in a text based on their roles in a sentence?

- (a) POS tagging
- (b) Text normalization
- (c) Stemming
- (d) Tokenization

15- When developing an application that involves extracting insights, sentiment, and key phrases from text data, which Azure service provides text analytics capabilities for language detection, sentiment analysis, and named entity recognition?

- (a) Azure Content Moderator
- (b) Azure Speech Service
- (c) Azure Text Analytics
- (d) Azure Translator

16- Which Pandas plotting method is used to create a horizontal bar plot?

- (a) df.plot.hist()
- (b) df.plot.barh()
- (c) df.plot.bar()
- (d) df.plot.line()

17- In supervised machine learning, which component represents the function that the model is trying to learn and approximate?

- (a) Optimizer
- (b) Hypothesis
- (c) Cost function
- (d) Labeled data

18- In machine learning, which algorithm creates a tree-like model of decisions and their possible consequences, where each internal node represents a decision based on an attribute, and each leaf node represents an outcome?

- (a) Linear Regression
- (b) Decision Tree
- (c) K-Means Clustering
- (d) Support Vector Machine (SVM)

19- Which service can you use to analyze images and extract text from them in Microsoft Azure?

- (a) Anomaly Detector
- (b) Personalizer
- (c) Metrics Advisor
- (d) Content Moderator

20- Which Numpy function is used to perform element-wise addition of two arrays?

- (a) `numpy.add()`
- (b) `numpy.append()`
- (c) `numpy.sum()`
- (d) `numpy.plus()`

__The End__



Government of Pakistan
National Vocational & Technical Training Commission (NAVTTTC)

Plot No, 38 Kirthar Rd, H-9/4 , Islamabad,



Roll no: 1153550

Name: Saira Saleem

CNIC: 3110178630484

Trade: Artificial Intelligence (Machine Learning; Deep Learning; Communication)

Test Date & Time: 22/05/2026 12:00 AM

Test Center Name: DevCastle Training institute |
Software house Bahawalnagar Campus

Examiner Name: Sohail Murtaza /

Signatures

Candidate Signature

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Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- In object-oriented programming, what is an abstract method?

- (a) A method that can't be called
- (b) A method that is automatically generated
- (c) A method that has no implementation
- (d) A method that can't be overridden

2- Which type of data includes non-numerical information such as labels or categories?

- (a) Numerical data
- (b) Qualitative data
- (c) Categorical data
- (d) Quantitative data

3- In machine learning, what is the process of selecting a subset of relevant features from the original set of features to improve model performance and reduce complexity?

- (a) Feature extraction
- (b) Feature selection
- (c) NER tagging
- (d) Tokenization

4- In machine learning, which loss function is commonly used for classification tasks and measures the dissimilarity between predicted probabilities and actual class labels?

- (a) Mean Absolute Error
- (b) Mean Squared Error
- (c) Huber Loss
- (d) Cross-Entropy

5- Which measure quantifies the strength and direction of a linear relationship between two variables?

- (a) Mean
- (b) Correlation Coefficient
- (c) Standard Deviation
- (d) Variance

6- What is the main goal of machine learning?

- (a) To program computers to think like humans
- (b) To enable computers to learn from data and improve performance over time
- (c) To automate tasks without using data
- (d) To optimize computer hardware

7- What method is used to select elements from a Numpy array based on a condition?

- (a) conditional_selection()
- (b) filter()
- (c) logical_select()
- (d) where()

8- What type of data consists of numerical values that represent quantities?

- (a) Binary data
- (b) Categorical data
- (c) Qualitative data
- (d) Quantitative data

9- When planning a container deployment for your application, which technology allows you to package and distribute applications along with their dependencies, ensuring consistency and portability across different environments?

- (a) Docker Containers
- (b) Azure Functions
- (c) Kubernetes
- (d) Virtual Machines

10- What is type casting in Numpy used for?

- (a) Changing the shape of an array
- (b) Changing the data type of an array
- (c) Changing the dimensions of an array
- (d) Changing the elements of an array

11- Which measures assess the spread or dispersion of data points around the mean?

- (a) Mean and Standard Deviation
- (b) Mean and Median
- (c) Variance and Standard Deviation
- (d) Range and Mode

12- In neural networks, what graphical representation visualizes the sequence of mathematical operations performed during forward and backward passes, aiding in understanding and optimization of the learning process?

- (a) Loss Function
- (b) Gradient Descent
- (c) Activation Function
- (d) Computational Graph

13- Which data structure is commonly used in Pandas to store tabular data?

- (a) Arrays
- (b) Lists
- (c) DataFrames
- (d) Sets

14- Which command is used to display the contents of a directory in the terminal?

- (a) ls
- (b) display
- (c) dir
- (d) show

15- In Pandas, which type of join combines rows from two DataFrames based on common columns, keeping only the matching rows?

- (a) Left join
- (b) Right join
- (c) Inner join
- (d) Outer join

16- If you're building a speech solution that involves converting spoken language into text, performing text-to-speech synthesis, and enabling speaker recognition, which Azure Cognitive Services offering is best suited for these tasks?

- (a) Azure Text Analytics
- (b) Azure Language Understanding
- (c) Azure Speech Service
- (d) Azure Translator

17- In Linux, which command is used to update the package list?

- (a) apt-get
- (b) apt-upgrade
- (c) apt-list
- (d) apt-update

18- In logistic regression for multiclass classification, what type of decision boundary is typically learned by the model?

- (a) Exponential decision boundary
- (b) Linear decision boundary
- (c) Non-linear decision boundary
- (d) Constant decision boundary

19- If you're developing an AI solution that involves extracting meaningful insights from unstructured text data, identifying sentiment, entities, and key phrases, which cloud service offers a comprehensive Text Analytics API?

- (a) Azure Cognitive Services
- (b) IBM Watson
- (c) Google Cloud AI
- (d) Amazon SageMaker

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- (a) numpy.add()
- (b) numpy.append()
- (c) numpy.sum()
- (d) numpy.plus()

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